INCH-POUND

A-A-50181A
December 15, 1993
SUPERSEDING
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October 30, 1988

COMMERCIAL ITEM DESCRIPTION

FRYERS, DEEP-FAT (GAS, HEAVY DUTY)

The General Services Administration has authorized the use of this Commercial Item Description in preference to S-F-700.

1. ABSTRACT

1.1 Abstract. This commercial item description covers commercial, heavy duty, floor mounted, gas heated, deep-fat fryers in the following sizes:

Size 1 - 30 pounds minimum production capacity
Size 2 - 60 pounds minimum production capacity
Size 3 - 90 pounds minimum production capacity
Size 4 - 125 pounds minimum production capacity

1.2 CID based part identification number (PIN). A document based PIN to identify sizes of items is included in section 5. This identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor except as information on "Government unique" data nameplates (see 5.4).

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The fryers shall have an enclosed cabinet designed for floor use equipped with 6 inch minimum adjustable stainless steel legs. The cabinet front, door, sides, and splash guard shall be fabricated of 300 series stainless steel conforming to ASTM A 167 or

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AMSC N/A FSC 7310

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400 series stainless steel conforming to ASTM A 176. The back and bottom shall be fabricated of stainless steel, aluminized steel, galvanized steel, or steel treated to resist corrosion. The cooking vessels shall be stainless steel and have a hand-operated drain valve. A stainless steel or nickel plated steel fat draining vessel with carrying handle(s), having a minimum capacity of 35 percent of the cooking vessels capacity, shall be furnished with each fryer. Each fryer shall be equipped with one large and two half baskets, constructed of stainless steel or nickel plated steel and a stainless steel cooking vessel cover with a handle. The fryer shall be equipped with a thermostat, an on-off switch, a power on light, and a signal light indicating burner(s) is on and up to selected temperature.

2.2 Gas consumptions and dimensions. The fryer dimension shall be within the dimensional limits specified in table I:

TABLE I. Gas consumption and dimensions

	Production capacity 1/ lbs/hr (minimum)	Btu/hr (minimum)	Dimensions (inches) (maximum)		
			Height 2/	Width	Depth
Size 1	30	60,000	37-1/2	14-1/2	30
Size 2	60	90,000	37-1/2	20	38
Size 3	90	110,000	37-1/2	21-1/2	38
Size 4	125	125,000	37-1/2	21-1/2	38

1/ Production capacity of raw to done 1/2 inch by 1/2 inch french fries that can be processed in one hour.
2/ Height from floor to top of cooking vessel including legs.

2.3 Fuel system. The fuel system shall be designed to burn natural gas and convert to burn propane or liquid petroleum gas (LPG) and comply with the Btu for the fryer size as specified in table I. The fryer shall be equipped with a pressure regulator, automatic pilot with safety shut-off, and an automatic ignition system. The pressure regulating valve shall conform to ANSI Z21.18. An American Gas Association (AGA) or Underwriters Laboratory (UL) approved quick disconnect (socket and plug) conforming to ANSI Z21.41 and a 5 foot long minimum flexible metal connector conforming to ANSI Z21.45 with a male pipe thread fitting on one end and a union with a female pipe fitting on the opposite end shall be provided (see 3.2). Fryers shall be supplied to burn natural gas. Orifices for LPG and instructions for conversion from natural gas to LPG and a pressure regulator shall be provided with each fryer.

2.4 Electrical requirements. Electrical controls shall operate on a nominal 120 volt, 60 hertz power source. The fryer shall be equipped with a 6 foot long minimum three-wire power supply cord with a National Electrical Manufacturers Association (NEMA) 5-15P plug.

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2.5 Thermostat. The fryer shall be equipped with an adjustable thermostat with a range of 200°F/275°F to 375°F/400°F, and a fixed over-temperature thermostat to shut off the entire gas system when the fat reaches a high limit of 430°F to 460°F.

2.6 Standard compliance. The fryer shall comply with American National Standards Institute (ANSI) Z83.13 and NSF International Standard No. 4 (see 3.2).

2.7 Data nameplate ("Government unique" requirement). The fryers shall be furnished with a data nameplate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data nameplate shall be made of minimum 20 gauge corrosion-resisting metal and attached to the fryer by rivets, screws, or welding in such a manner as to meet the applicable NSF International sanitary requirements for this equipment. In addition to the marking requirements of ANSI Z83.13, the plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

Contract Number
National Stock Number
Procurement Instrument Identification Number
CID Based Part Identification Number (PIN)
Btu Rating
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List Only if Different from Manufacturer)
Manufacturer's Model and Serial Number
Government Approved Manual Number (see 5.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the fryer.

2.8 Instruction plate. Instruction plates made of corrosion-resisting metal shall be attached to the fryer. The instruction plates shall have the wiring diagram, instructions for start-up, operation, shutdown, and preventative maintenance procedures. In addition, information and markings as required by ANSI Z83.13 shall also appear on the plate. Lettering shall not be less than 1/16 inch high.

2.9 Service and maintenance. All major assemblies and installed components shall be accessible for maintenance, repair, service, and replacement with minimum disturbance to adjacent components in fryer and without usage of special tools.

2.10 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables

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contained in the latest revision of Federal Standard No. 376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

2.11 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description unless otherwise specified.

2.12 Workmanship. The finished fryers shall be complete, clean, and free of fabrication marks, scratches, dents, sharp edges, sharp corners, burrs, slivers, or visible evidence of rust. Rivets, when used, shall be tight and the rivet heads and peened ends shall be round and concentric and in full contact with the joined metal surfaces. There shall be no empty rivet holes. Welds shall be clean, continuous, sound, and free of flux, pits, buildups, undercuts, and shall be ground so as to blend with the adjacent surfaces. Spot welds shall be free from pits, burn throughs, and flashes.

3. QUALITY ASSURANCE

3.1 Contractor certification. The contractor shall certify that the product offered meets the salient characteristics and requirements of this commercial item description and except for those Government unique requirements specified, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace or that had successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the fryer conforms to the requirements of NSF International Standard No. 4 (see 2.6), NEMA standards (see 2.4), and ANSI Z83.13 (see 2.6), and that the applicable components conform to ANSI Z21.18, ANSI Z21.41, and ANSI Z21.45 (see 2.3), as applicable.

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3.2.1 American National Standards Institute (ANSI). Acceptable evidence of meeting the applicable requirements of ANSI Z21.18, ANSI Z21.41, ANSI Z21.45, and ANSI Z83.13 shall be a photostatic copy of the American Gas Association (AGA) Appliance Certificate; a listing of the fryers, regular and quick disconnect devices and connections in the AGA Directory of Certified Appliances and Accessories.

3.2.2 NSF International. Acceptable evidence of meeting the requirements of NSF International Standard No. 4 shall be a listing in the current edition of the NSF International "Listing of Food Service Equipment" and display of the NSF International mark on the finished product, or a certified test report from a recognized independent laboratory acceptable to the Government indicating compliance with the applicable NSF International requirement(s).

3.2.3 National Electrical Manufacturers Association (NEMA). The contractor shall submit to the contracting officer a written certificate of compliance as proof that the fryers meet the applicable requirements of NEMA.

4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export, as specified (see 5.2).

4.2 Commercial.

4.2.1 Commercial packaging. Each fryer shall be cleaned, preserved, and packaged in accordance with ASTM D 3951.

4.2.2 Commercial packing. The fryer shall be packed in accordance with ASTM D 3951.

4.3 Export.

4.3.1 Export packaging. Each fryer shall be disassembled to the extent determined by the contractor to protect parts susceptible to damage during shipment and storage and for reduction of cubage. Parts removed shall be packaged in a domestic fiberboard box constructed and closed in accordance with PPP-B-636. Parts not removed shall be secured in place with cushioning or pressure-sensitive tape. Doors and drawers shall be secured in a closed position with filament reinforced pressure-sensitive tape. The fryer and parts shall be cleaned, preserved, and cushioned to afford protection against corrosion, deterioration, and physical damage during shipment and storage.

4.3.2 Export packing. Each fryer shall be packed in a snug-fitting wood-cleated, plywood box, conforming to domestic type, style A or J, grade B, type 2 load of PPP-B-601 or mailed wood box conforming to class 1, style 2 or 4, grade B, type 2 load of PPP-B-621. The contents of each box shall be blocked and braced to prevent movement while the box is in transit. When legs or casters are not disassembled, the fryer shall be blocked in a manner

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to provide approximately 1 inch clearance between the legs or casters and the bottom of the box. Each shipping container shall be closed and strapped in accordance with the appendix of PPP-B-601 and PPP-B-621, as applicable.

4.4 Marking (commercial and export). Marking of unit pack and shipping container shall be in accordance with MIL-STD-129.

5. NOTES

5.1 Intended use. The fryers covered by this commercial item description are intended for use in kitchens or galleys for deep fat frying of food products.

5.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this commercial item description.
b. PIN designating size required (see 1.1, 1.2, and 5.4).
c. Government approved manual number to be included on data name plate (see 2.7).
d. Type of packaging and packing required (see 4.1).

5.3 Sources for documents.

5.3.1 Sources for non-Government association documents.

American Gas Association (AGA)
8501 East Pleasant Valley Road
Cleveland, OH 44131

ANSI Z21.18 - Gas Appliance Pressure Regulators
ANSI Z21.41 - Quick-Disconnect Devices for Use with Gas Fuel
ANSI Z21.45 - Flexible Connectors of Other Than All-Metal Construction for Gas Appliances
ANSI Z83.13 - Gas Food Service Equipment (Deep Fat Fryers)

ANSI documents are available from:

American National Standards Institute (ANSI)
11 West 42nd Street
New York, NY 10036

ASTM A 167 - Stainless and Heat Resisting, Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM A 176 - Stainless and Heat Resisting, Chromium Steel Plate, Sheet, and Strip
ASTM D 3951 - Standard Practice for Commercial Packaging

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ASTM documents are available from:

American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

NEMA documents are available from:

National Electrical Manufacturers Association (NEMA)
2101 L Street, NW
Washington, DC 20037

NSF International Standard No. 4 - Commercial Cooking Rethermalization and Powered Hot Food Holding and Transport Equipment

NSF International documents are available from:

NSF International
3475 Plymouth Road
P.O. Box 130140
Ann Arbor, MI 48113-0140

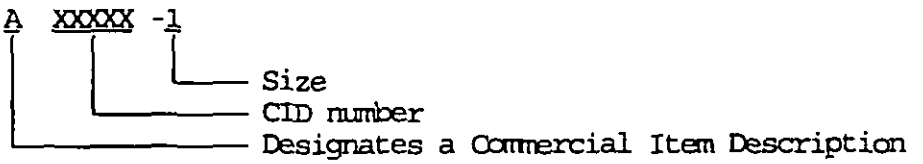
UL standards are available from:

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062

5.3.2 Source for Government documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

5.4 Part identification number (PIN). The PIN's to be used for items required by this CID are created as follows:



5.5 "Government unique" requirements. Whenever a "Government unique" requirement is included in the title of a paragraph under "Salient Characteristics", it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.

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A-A-50181A

MILITARY INTERESTS:

Custodians

Army - GL
Navy - SA
Air Force - 99

Review Activities

Army - MD, QM
Navy - MC, MS
Air Force - 50, 84
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

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